

Towards a mechanistic understanding of the drivers of plant-pollinator interactions

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Abstract

Flowering plants exhibit an astonishing diversity of reproductive structures, which shapes how they interact with their pollinators. However, to what extent reproductive traits or other drivers like phenology and species abundances shape plant-pollinator interactions is still an open question. To disentangle the relative effects of species phenology, abundances and traits in plant-pollinator interactions, we compared pollinator visitation rates, diet breadth, specialization from plant species that vary in reproductive traits, phenology and abundances. To

do this, we observed plant-pollinator interactions for a whole flowering season in three different botanical gardens located in an urban setting. We anticipate that the ecological role of the different plant and pollinator species are largely driven by their phenological overlap and abundance, and only certain taxonomic groups are strongly influenced by trait matching processes. Our results highlight the relevance of considering multiple mechanisms to understand plant-pollinator interactions.

Introduction

Flowering plants, pollinators, and their interactions form complex networks that are fundamental to the architecture of biodiversity. The structure and composition of these networks exhibit strong spatio-temporal variation, shaped by species attributes and environmental conditions (Burkle & Alarcón, 2011; Tylianakis & Morris, 2017). While ecological networks provide insights into the ecological processes that govern community assembly (Bascompte & Jordano, 2007; Guimaraes Jr, 2020), documenting interactions remains extremely labour-intensive (Chacoff et al., 2012). Moreover, the resulting network structures are also highly dependent on sampling effort and methodology (Jordano, 2016). At the same time, interactions are often summarised into single metrics and analysed with a diverse array of statistical methods, each offering potentially distinct perspectives on the ecological system (Delmas et al., 2019; Peralta et al., 2024). Despite pronounced temporal dynamics, networks are often treated as static entities (CaraDonna et al., 2017; Poisot et al., 2015), requiring the aggregation or simplification of relevant information when evaluating ecological patterns. The complexity of plant-pollinator networks, together with the limitations of current analytical practices, can obscure rather than clarify our view of ecological systems. Addressing these challenges requires systematic approaches that integrate multiple layers of network complexity to improve both understanding and predictive power (Blüthgen & Staab, 2024; Dormann et al., 2017).

There is strong evidence that species phenology, abundances, and traits are key ecological drivers shaping species interactions and network structure (Kaiser-Bunbury et al., 2014; Vázquez et al., 2009), yet most studies lack the appropriate data to assess their combined influence (Olito & Fox, 2015; Vázquez et al., 2009). Phenology is considered one of the main determinants of network structure, as temporal overlap between species constrains the pool of potential interactions (Benadi et al., 2014; Encinas-Viso et al., 2012), with approximately one-third of all potential interactions estimated to be impossible due to phenological mismatches (Olesen et al., 2011). When two species overlap in time, the probability that they interact is generally considered a function of their relative abundances and species traits (Olito & Fox, 2015; Stang et al., 2007), along with other abiotic and biotic factors (e.g. temperature and species competition). Abundance influences both species' roles (Fort et al., 2016; Simmons et al., 2019) and overall network structure (Canard et al., 2014; Vázquez et al., 2009), but independent estimates are rarely available and are often inferred from interaction data (Blüthgen & Staab, 2024).

Traits, in turn, provide mechanistic insights, as they constrain the range of possible partners and help define species' ecological roles within networks (Bartomeus et al., 2016; Coux et al., 2016; Dehling et al., 2016). For instance, interaction probability often peaks when pollinator proboscis length matches floral tube depth (Klumpers et al., 2019), yet imperfect matches can still allow interactions. Although multiple traits often act simultaneously, capturing their multidimensional nature remains challenging (Eklöf et al., 2013; Lanuza et al., 2023), though phylogenetic approaches may help integrate such complexity. Further, most pollinators tend to be generalist, and the relative importance of traits is expected to decline with increasing generalism (Peralta et al., 2020). The strong context-dependency of these different ecological drivers constrains predictive power and highlights the need for high-quality data to evaluate multiple scenarios across taxonomic groups and spatiotemporal scales.

Recent evidence suggests that distinct temporal scales can provide different views of the structure and processes that shape plant-pollinator interactions (Miele et al., 2020; Schwarz et al., 2020). Species and interactions are highly dynamic, shifting markedly between years (Chacoff et al., 2018; Petanidou et al., 2008), across seasons (Souza et al., 2018; Vizentin-Bugoni et al., 2016), and even within days (Schwarz et al., 2021). At fine scales, interactions are shaped by physiological responses to abiotic factors (e.g., temperature, light, precipitation) that determine activity periods, whereas variation across days or weeks also reflects phenological succession and changes in the species pool (Bramon Mora et al., 2020; CaraDonna et al., 2017). Although some of this variation arises from natural dynamics, global change is expected to disrupt and intensify temporal fluctuations (Duchenne et al., 2020; Kharouba et al., 2018). While aggregated snapshots of communities are valuable for describing structural and ecological properties at broad scales (Bascompte & Jordano, 2007), the relative impact of ecological drivers may vary with temporal scale, underscoring the need to account for temporal dynamics to fully understand plant-pollinator interactions.

Although there is no consensus on the range of ecological tools used to analyze ecological networks, combining approaches can provide complementary insights into predicting species interactions (Delmas et al., 2019). For example, structural metrics are often compared with synthetic networks or null models built under different ecological constraints to test for significance, or across empirical networks to assess variation among systems (Marjakangas et al., 2022; Santamaria & Rodriguez-Girones, 2007; Vázquez & Aizen, 2004), a strategy widely applied to evaluate properties such as nestedness (Bascompte et al., 2003; Payrató-Borras et al., 2019; ?) and modularity (Dormann et al., 2013; Olesen et al., 2007). In addition, statistical approaches can complement structural analyses by explicitly assessing the influence of biotic and abiotic drivers. Common approaches include matrix comparisons such as Mantel or Procrustes tests [Burkle & Alarcón (2011); REF2; REF3] and conventional models such as generalized linear models [REFs]. In addition, complex associations that involve direct and indirect effects have traditionally been analyzed using structural equation models (e.g., Kaiser-Bunbury et al., 2014; Lázaro et al., 2020; Theodorou et al., 2017).

Regression models enable more refined evaluations of the impact of biotic and abiotic drivers on species interactions or network properties (e.g., Geslin et al., 2013; Herrera, 2020; Maglianesi

et al., 2024), while also allowing ecological predictions [e.g. Crea et al. (2016); REFs]. More complex associations that involve both direct and indirect effects have traditionally been analyzed using structural equation models, applied across the different variables of interest (e.g., Kaiser-Bunbury et al., 2014; Lázaro et al., 2020; Theodorou et al., 2017). Recent attempts to improve the predictive power of species interactions have tried to overcome the limitations of conventional analytical methods like the non-independence of observations and the potential non-linear effects of key variables, while incorporating important drivers such as traits, abundances, and phylogenetic relationships allowing predictions of interaction probability [i.e. tapnet approach; Benadi et al. (2022)]. Finally, flexible machine learning methods have emerged as particularly promising for enhancing predictive accuracy (Strydom et al., 2021). For instance, Pichler et al. (2020) found that flexible machine learning models outperformed traditional regression models in predicting species interactions in plant-pollinator networks. However, multiple analytical approaches exist to evaluate and predict ecological interactions, and skepticism remains due to the varying implementations of these methods, which strongly depend on sampling methods and effort (Blüthgen & Staab, 2024; Dormann et al., 2025). Different analytical approaches can be complementary, each providing unique and relevant insights to predict species interactions (Peralta et al., 2024).

In this study, we aim to evaluate the relative importance of biotic and abiotic drivers of plant-pollinator interactions within seasonal dynamics using various analytical methods, and to assess the predictive power of species interactions under different analytical approaches and temporal scales (Figure 1). To achieve this, we collected data on plant-pollinator interactions, along with associated biotic (i.e., abundances, traits, and phenology) and abiotic variables (i.e., temperature), in three German botanical gardens. Our objective is to contribute to a mechanistic understanding of these drivers by using a semi-controlled environment with numerous co-flowering species that exhibit heterogeneous traits and diverse phylogenetic backgrounds. This setting allows us to test multiple potential combinations of pairwise interactions within the gardens and to explore in depth the role of traits and phylogenetic relatedness as surrogates for unmeasured traits.

Methods

Study site and sampling method

Plant-pollinator interactions were monitored during the entire flowering season of 2023 in the botanical gardens of Halle, Jena, and Leipzig (Germany). These gardens belong to PhenObs initiative (Nordt et al., 2021), which consists of a network of botanical gardens across the Northern Hemisphere that records phenological observations on a large set of herbaceous species. The objective of PhenObs is to evaluate the impacts of climate change on species performance (e.g., Sporbert et al., 2022). In this study, plant-pollinator interactions were recorded for a subset of species from the PhenObs project within these three gardens. Each garden was visited weekly, and plant-pollinator interactions were sampled during 15 mins on each focal

plant. On each sampling day, the different focal plants were sampled in a random order and only pollinator individuals that contacted the reproductive organs of the plant were recorded as interacting species. In addition, to create a comprehensive inventory of pollinators and their interactions within the gardens, 10 random sites were selected each sampling day, and plant-pollinator interactions were monitored for three minutes at each random site, resulting in a total of 30 minutes per sampling day and garden. Therefore, two sampling methodologies were implemented, one that recorded interactions on PhenObs plants, and another that consisted of random stops within the garden, with the specific goal of describing all potential pollinators present in the garden.

Sampling effort and plant taxonomic coverage

This study monitored a total of 78 focal plant species that belonged to PhenObs initiative, which accounted for 68 different genera, 33 families and 22 different orders. In total, we sampled pollinators on 46 plants species in Halle (52.3 hours), 43 in Jena (41.3 hours) and 47 in Leipzig (49.8 hours), and only 12 species were shared across the three gardens. In addition, with the random stops across the garden we sampled a total of XXXX species that comprised XXXX genera, XXXXX families and XXXXX orders. This study accounted for a total of 134 plant species in Halle (10.1 hours), 124 in Jena (11.3 hours), and 125 in Leipzig (10.4 hours). Explain accumulation curves here!!!

Data description

To evaluate the different mechanisms that drive plant-pollinator interactions, detailed information on abundance, phenology and species traits for both plants and pollinators was recorded as follows:

- (i) Floral abundance was annotated for each focal plant species, and given that botanical gardens are a semi-control setting with relatively low abundance of pollinators, we were also able to annotate the different number of pollinator individuals on each sampled plant when documenting interactions. Despite not being fully independent of interactions, this approach offers a more realistic estimate of pollinator abundance than the standard method of estimating abundances from interactions, which can bias our understanding of the mechanisms driving network structure (Blüthgen et al., 2006; Vizentin-Bugoni et al., 2014). Note that flowering species with tiny densely packed flowers were considered at inflorescence level (15.2% of the total flowering species considered in this study).
- (ii) Given that plant phenology was already monitored on the PhenObs project, we have weekly phenological records with flowering percent for all focal plants. For 10 species with missing phenology data, we conducted generative additive models (GAM) with our records of flower number to build the phenology on the same sampling dates as the other species by using the *mgcv* package. Unlike plants, pollinator phenology relies

on ‘stochastic’ interaction observations that may not represent their complete flying period. For this reason, we opted to complement our observational data from the Global Biodiversity Information Facility (GBIF). Germany is one of the countries with highest records of occurrence data (Rocha-Ortega et al., 2021), and in particular, their cities (e.g., botanical gardens) contain most of those records (Sweet et al., 2022). Thus, we opted to extract occurrence records around the three cities for the different pollinator species to complement the different pollinator phenologies. We first created a centroid within the 3 cities, and then, created a rectangular shape around it that expanded approximately 1000 kms in longitude and 500 kms in latitude (Figure S1). We used this shape because we expect environmental variables to be more consistent and phenology to be more similar across a longitudinal range than a latitudinal one. Thus, our download resulted in 2,933,772 occurrence records (<https://doi.org/10.15468/dl.k5cnuf>).

- (iii) Plant reproductive traits were selected following the main traits driving the trait spectrum of flowering plants as shown in Lanuza et al. (2023) (i.e., flower number, plant height, style length, flower size, ovule number and selfing). In addition, we also included phenological traits (i.e., flower lifespan and flowering length), floral rewards (nectar volume and number of pollen grains per flower) and seed weight as a measurement of the reproductive investment. Pollinator traits included intertegular distance, body length and proboscis length. Both intertegular distance and body length were measured from images using ImageJ, and proboscis length was extrapolated for bee species by using the pollimetry package (Kendall et al., 2019). Non-bee species were complemented with proboscis values from the literature and images when possible.

Plant and pollinator phylogeny

*Phylogeny... Plant phylogeny is easy, bee phylogeny too...

*compute phenology matrices: presence absence and probability ones based by calculating the probability of observing both if they are independent events divided by total number of sampling days

CONCEPTUAL GRAPH. **Figure 1**

Analysess

To unravel the role of time in our understanding of the ecological drivers of plant-pollinator networks, this study analyses the relative importance of abundance, phenology and traits at the whole season level, by dividing in early-mid-and late season and by analysing at the sampling day level.

Results

PROVIDE: Percentage of forbidden links due to phenological mismatches by garden.

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Supporting information

Figure S1: Some sort of combination of the code below and the graph of location from the poster